



$E[(Y - f(x))^2] = \text{bias}^2 + \text{variance} + \text{irreducible error}$
 As model flexibility increases, bias decreases while variance increases, leading to a U-shaped test error curve.

(b) As model flexibility increases, squared bias decreases because the model can better approximate the true underlying function, while variance increases due to greater sensitivity to the training data. Training error decreases monotonically as more flexible models can fit the training data more closely. Test error exhibits a U-shaped curve since it reflects the trade-off between bias and variance. The irreducible (Bayes) error remains constant because it is inherent in the data-generating process.

2. Advantages of a very flexible approach:

A flexible model can capture complex and nonlinear relationships between predictors and the response. As a result, it has low bias and can achieve low training error when the true underlying relationship is highly complex. Flexible methods are more capable of approximating it accurately.

Disadvantages of a very flexible approach:

Very flexible methods tend to have high variance, meaning they are sensitive to the specific training data used. This can lead to overfitting, where the model fits noise rather than the true signal, resulting in poor generalization performance on new data. Flexible models are also often harder to interpret and may require more data to perform well.

Advantages of a less flexible approach: Less flexible models usually have lower variance and are more stable across different training samples. They are less prone to overfitting, easier to interpret, and often perform well when the true relationship is simple or approximately linear.

Disadvantages of a less flexible approach: The main disadvantage is high bias. If the true relationship is complex, an inflexible model may underfit the data and fail to capture important patterns, leading to higher test error.

A more flexible approach is preferred when the true relationship between predictors and response is highly nonlinear or complex. The sample size is large relative to the noise level, prediction accuracy is more important than interpretability.

A less flexible approach is preferred when the true relationship is simple or approximately linear, the sample size is small or the noise level is high. Interpretability and model stability are important.

3. In unsupervised learning, overfitting means that the model captures noise or idiosyncratic structure in the data rather than meaningful underlying patterns, resulting in representations or structures that do not generalize to new data.

Example of dimension reduction model: Using too many principal components in PCA. In PCA, if too many principal components are retained, the model will not only capture the main directions of variation in the data but will also start to capture noise. In this case, the dimension reduction model can reconstruct the original data almost perfectly on the training data but performs poorly on new data, indicating that the model has overfitted the noise.

Example of clustering model: Choosing a value of k that is too large in k -means. In clustering, if the number of clusters k is chosen too large, the model will split data points that should belong to the same cluster into many small clusters, or even create noise as independent clusters. These clusters are not stable and cannot be reproduced with new data.

Example of generative models: GMM using too many mixture components. In generative models, if the number of mixture components is too large, the model may assign components to small numbers of samples or even a single sample. The likelihood of the training data is very high. However, the newly generated samples look almost identical to the training data, lacking diversity. This indicates that the model is not learning the data distribution but rather memorizing the training samples.

$$4. (1) \because B_3 \sim N(0, 5) \quad \therefore P(B_3 \geq \frac{1}{2}) = 1 - P(B_3 < \frac{1}{2}) =$$

$$1 - \Phi\left(\frac{\frac{1}{2}}{\sqrt{5}}\right) \approx 1 - \Phi(0.2236) \approx 0.588$$

$$(2) B_3 > B_1 + 2 \Leftrightarrow B_3 - B_1 > 2 \quad \because B_1 \perp (B_3 - B_1)$$

$$\therefore P(B_1 \leq \frac{1}{2}, B_3 > B_1 + 2) = P(B_1 \leq \frac{1}{2}, B_3 - B_1 > 2) = P(B_1 \leq \frac{1}{2}) \times$$

$$P(B_3 - B_1 > 2) \quad \because B_1 \sim N(0, 1), B_3 - B_1 \sim N(0, 2)$$

$$\therefore P(B_1 \leq \frac{1}{2}) = \Phi(0.5), P(B_3 - B_1 > 2) = 1 - \Phi\left(\frac{2}{\sqrt{2}}\right)$$

$$\therefore P(B_1 \leq \frac{1}{2}, B_3 > B_1 + 2) = \Phi(0.5) \cdot [1 - \Phi\left(\frac{2}{\sqrt{2}}\right)] \approx 0.084$$

$$(3) E = \int_{0.5}^{0.75} \int_{0.5}^{0.75} P_0 \, dx \, dy$$

$$P(E) = P\left\{\max_{0.5 \leq x \leq 0.75} B_0 \leq 6\right\} = 2 \Phi\left(\frac{6}{\sqrt{0.5}}\right) - 1 \approx 2 \times 0.9711 - 1 = 0.942$$

$$(4) B_3 \sim N(0, 2), B_4 - B_3 \sim N(0, 2) \quad \therefore B_4 - B_3 \perp B_3$$

$$\therefore B_4 = (B_4 - B_3) + B_3, P(B_4 \leq 0 | B_3 \geq 0) = P(B_4 - B_3 \leq -B_3 | B_3 \geq 0)$$

$$\text{Let } X = B_3, Y = B_4 - B_3 \quad \text{then } X \sim N(0, 2), Y \sim N(0, 2)$$

$$X \perp Y, P(B_4 - B_3 \leq -B_3 | B_3 \geq 0) = P(X + Y \leq 0 | X \geq 0) =$$

$$P(Y \leq -X | X \geq 0) = \frac{P(Y \leq -X \cap X \geq 0)}{P(X \geq 0)} = \frac{45^\circ}{180^\circ} = \frac{1}{4}$$

5. (a) Let $\{B_t : t \in [0, 1]\}$ be a standard Brownian motion.

Consider the uniform partition $t_k = \frac{k}{n}, k = 0, 1, 2, \dots, n$. Let L_n be the length of the polygonal path connecting the points (t_k, B_{t_k}) .

For each segment, $\sqrt{(t_k - t_{k-1})^2 + (B_{t_k} - B_{t_{k-1}})^2} \geq |B_{t_k} - B_{t_{k-1}}|$.

Hence $L_n \geq \sum_{k=1}^n |B_{t_k} - B_{t_{k-1}}|$. By independent increments, $B_k^2 - B_{k-1}^2 \sim N(0, \frac{1}{n})$.

Since the increments are independent, therefore

$$|B_k^2 - B_{k-1}^2| \leq \frac{1}{\sqrt{n}} |Z_k|, Z_k \sim N(0, 1), i, j, 1, 2$$

$$\therefore \sum_{k=1}^n |B_k^2 - B_{k-1}^2| \leq \frac{1}{\sqrt{n}} \sum_{k=1}^n |Z_k| = \frac{1}{\sqrt{n}} \left(\sum_{k=1}^n |Z_k|^2 \right)^{1/2}$$

By the strong law of large numbers, $\frac{1}{n} \sum_{k=1}^n |Z_k|^2 \xrightarrow{a.s.} E[Z^2] = \frac{1}{2} > 0$

$$\therefore \sum_{k=1}^n |B_k^2 - B_{k-1}^2| \xrightarrow{a.s.} 0$$

Since $L_n \geq \sum_{k=1}^n |B_k^2 - B_{k-1}^2| \xrightarrow{a.s.} 0$, we conclude that the

length of the graph of Brownian motion on $[0, 1]$ is infinite with

probability 1.

5. (b) Let $\{B_t : t \in [0, 1]\}$ be standard Brownian motion. For $t \in [0, 1]$

(1) Choose a sequence of disjoint small intervals near t .

Assume first $t \in (0, 1)$. Define $s_n := t + 2^{-n}, n \geq 1$, then $s_n \downarrow t$.

Consider the increments on disjoint intervals: $\Delta_n := B_{s_n} - B_{s_{n-1}}$.

The intervals $[s_{n-1}, s_n]$ are pairwise disjoint, hence by independent increments the random variables Δ_n are independent.

$\Delta_n \sim N(0, s_n - s_{n-1}) = N(0, 2^{-n})$.

Define the local slope on the small interval: $\hat{A}_n := \frac{B_{s_n} - B_{s_{n-1}}}{s_n - s_{n-1}}$.

$\hat{A}_n \sim N(0, 2^{-n(n-1)}) \therefore \hat{A}_n \sim N(0, 2^{-n(n-1)})$.

(2) Show the slope $\hat{A}_n$ blow up.

Let $A_n := \{|\hat{A}_n| \leq n\}$ write $\hat{A}_n = 2^{-\frac{n(n-1)}{2}} Z_n$ where $Z_n \sim N(0, 1)$.

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$\therefore P(A_n) = P(|Z_n| \leq \frac{n}{2^{-\frac{n(n-1)}{2}}})$ Let $a_n := \frac{n}{2^{-\frac{n(n-1)}{2}}}$, $a_n \rightarrow 0$ exponentially.

fact, for small x , $P(|Z| \leq x) \leq Cx$ for some constant $C > 0$.

$C < \sqrt{\frac{2}{\pi}}$ $\therefore$ for large n , $P(A_n) \leq C a_n$ for some constant

$C > 0$. $\therefore \sum_{n=1}^{\infty} P(A_n) < \infty$. By the first Borel-Cantelli lemma,

$P(A_n \text{ i.o.}) = 0$ $\therefore$ with probability 1 there exists $N(\omega)$ such

that for all $n > N(\omega)$, $|\hat{A}_n| > n$, $\therefore |\hat{A}_n| \rightarrow \infty$ almost surely.

(3) Conclusion (from the above)

If B is differentiable at t , then the difference quotient

$\frac{B_t - B_s}{t - s}$ would converge to a finite limit as $s \downarrow t$ and hence

would be bounded in a neighborhood of t . However, $|\hat{A}_n| \rightarrow \infty$

almost surely. $\therefore$ There is a contradiction. $\therefore B$ is not

differentiable at t with probability 1. The argument is also

using a sequence approach 1 from the left. Hence, for any

$t \in [0, 1]$, $P(\text{Brownian motion is differentiable at } t) = 0$.